INCH-POUND

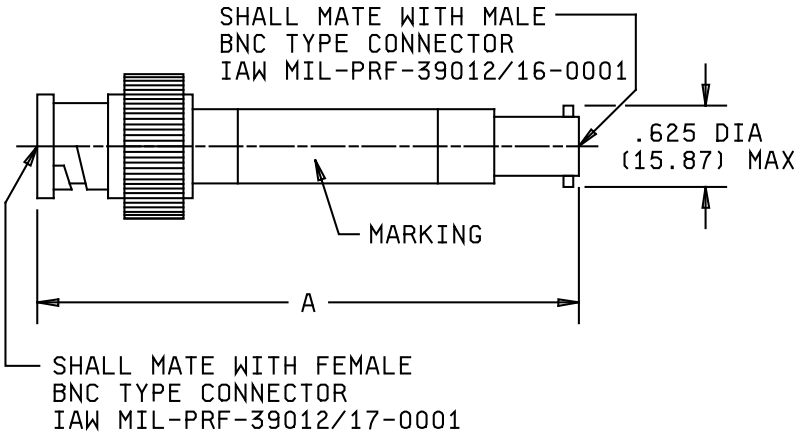
MIL-DTL-3933/19C
15 August 2011
SUPERSEDING
MIL-DTL-3933/19B
7 November 2002

DETAIL SPECIFICATION SHEET

ATTENUATORS, FIXED, SPACE LEVEL, NON SPACE LEVEL, COAXIAL LINE (BNC CONNECTORS), FREQUENCY RANGE: DC TO 4.5 GHZ, CLASS III, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Operating frequency range: DC to 4.5 GHz.

Test frequency range for SWR and attenuation measurements, measured or recorded, when using an Automatic Network Analyzer (ANA) or equivalent test equipment: 100 MHz (or lower) to 4.5 GHz.

Basic test frequency for sensitive measurements like connector repeatability, temperature sensitivity, power sensitivity and any other measurements where a 'delta' specification is imposed: 2.0 GHz.

Operating temperature range: -55°C to 125°C.

Maximum weight: See table I.

REQUIREMENTS:

Qualification: Not required.

First Article Testing shall be in accordance with the qualification inspection table of MIL-DTL-3933.

Dimensions and configuration: See figure 1 and table I.

Nominal impedance: 50 ohms.

Attenuation: See table I. Stability and sensitivity:

Maximum change in attenuation after:	**up to 10 dB, inclusive (dB)**	**over 10 dB (dB/dB)**
Temperature change or thermal shock	0.05	0.005
Vibration or shock	0.1	0.01
Moisture resistance or salt spray	0.2	0.02
Peak power	0.05	0.005

Frequency sensitivity, maximum: 0.1dB/dB/GHz.
Temperature sensitivity of attenuation, maximum: 0.0004 dB/dB/°C.

Power rating: See table I.
Power sensitivity for full input power, maximum: 0.005 dB/dB/Watt.

VSWR: See table I.

Connector repeatability, maximum variation in attenuation: 0.02 dB.

Part or Identifying Number (PIN): M3933/19- (and dash number from table I with three device level designators are space level (T), screened (S) and non-screened (N) options).

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TABLE I. Electrical characteristics.

Dash number		Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/				Weight (oz) Maximum	Dimension A (inches) Maximum
		Nominal	Deviation DC to 4.5 GHz	AV (W) 2/ (continuously)	Peak (kW) 3/	DC to 1.0 GHz	1.0 to 2.0 GHz	2.0 to 3.0 GHz	3.0 to 4.5 GHz		
01	NST	1	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
02	NST	2	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
03	NST	3	±0.3	4	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
04	NST	4	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
05	NST	5	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
06	NST	6	±0.3	3	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
07	NST	8	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
08	NST	10	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
09	NST	11	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
10	NST	12	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
11	NST	15	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)

See footnotes at end of table I.

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB		Maximum power input at 25°C		VSWR maximum 1/				Weight (oz) Maximum	Dimension A (inches) Maximum
	Nominal	Deviation DC to 4.5 GHz	AV (W) 2/ (continuously)	Peak (kW) 3/	DC to 1.0 GHz	1.0 to 2.0 GHz	2.0 to 3.0 GHz	3.0 to 4.5 GHz		
12 NST	20	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
13 NST	25	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
14 NST	30	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)
15 NST	40	±0.9	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
16 NST	50	±1.2	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
17 NST	60	±1.4	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	3.0 (76.2)
18 NST	13	±0.5	2	0.5	1.10:1	1.15:1	1.20:1	1.25:1	4	2.7 (68.6)

1/ VSWR value is for both ends.
2/ Power input is derated linearly to 0.5 watts at +125°C.
3/ Peak power for a duty cycle of 5×10^{-4}; maximum pulse duration of 5 microseconds.

Referenced documents. In addition to MIL-DTL-3933 this specification sheet references the following documents:

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MIL-PRF-39012/16
MIL-PRF-39012/17

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5985-2010-017)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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